

Machine learning for ecology and evolution research using natural history collections

a hands-on workshop

Organizers: Deborah L Paul, Matt Yoder, Species File Group, INHS PRI

Instructors: Bruno de Medeiros, Elizabeth Postema, The Field Museum

https://prairieresearchinstitute.github.io/2026_ml_eco_evo_collections_workshop/

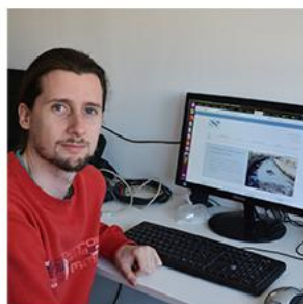
Hands-on Workshop: Machine learning for ecology and evolution research using natural history collections

Welcome to the gitHub repository for this ML/AI event hosted by the organizing team from the INHS Species File Group at PRI. We're pleased to bring guest instructors [Bruno de Medeiros](#), and [Elizabeth Postema](#), from the Field Museum to offer this hands-on workshop. Together, we will explore machine learning tools that can be used to accelerate research using natural history collections. We will focus on two specific domains: **custom-trained image classification and detection models**, and **using language models to extract structured data from inputs like pdfs or collection labels**. We will rely on user friendly tools to show how to train and deploy image models and how to interact programmatically with large language models. We will discuss pros and cons of cloud-based tools or those that can be run locally, and options for managing costs.

- Date: August 26, 2026
- Where (in person): Forbes 1005. [Google Maps](#), [Open Street Map](#)
- Time: 900 - 500 PM
- Instructors: Bruno de Medeiros, Assistant Curator of Insects, Field Museum and Elizabeth Postema, Post Doctoral Research Scientist, Field Museum
- Helpers: [Matt Yoder](#), [Geoff Ower](#), [Deborah Paul](#)



species file GROUP



Ed DeWalt
José Luis Pereira
Hernán Lucas Pereira
Phil Anders
Yuri Roskov
Lesley Deem
Heidi Hopkins



Dmitry Mozzherin
Matt Yoder
Dmitry Dmitriev
Deborah Paul
Geoff Ower
Tom Klein



Prairie Research Institute
Illinois Natural History Survey



Catalogue of Life



TaxonWorks

Setting the Tone – our SFG Code of Conduct

- Be kind
- Keep conversations
 - Constructive, respectful
 - Collegial
 - Cogent
 - Transparent
- Be open – a growth mindset
- See also [SFG Code of Conduct](#)



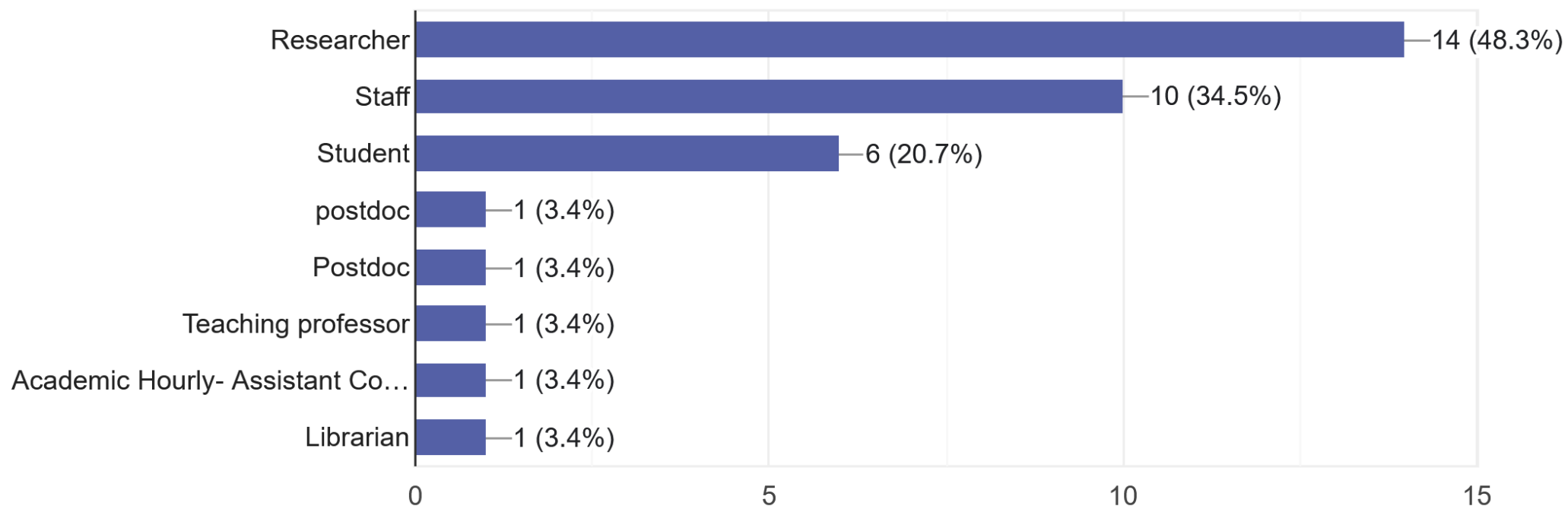
Schedule and Logistics

- 9:00 Welcome
- 9:15 training image models – Elizabeth
- Noonish – exact lunch time TBD
- Afternoon - using Ilms - Bruno
- 4:30 Wrap Up Conversation
- 5:00 End

Our cohort roles

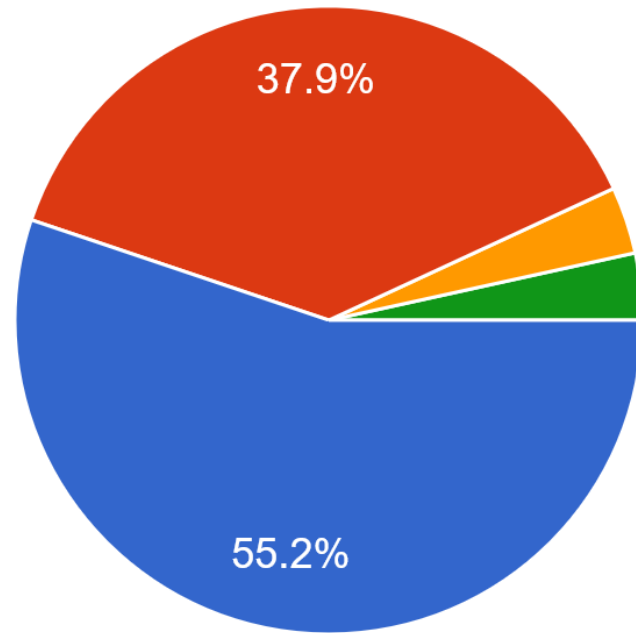
Role

29 responses



Which of the topics we're offering interest you the most?

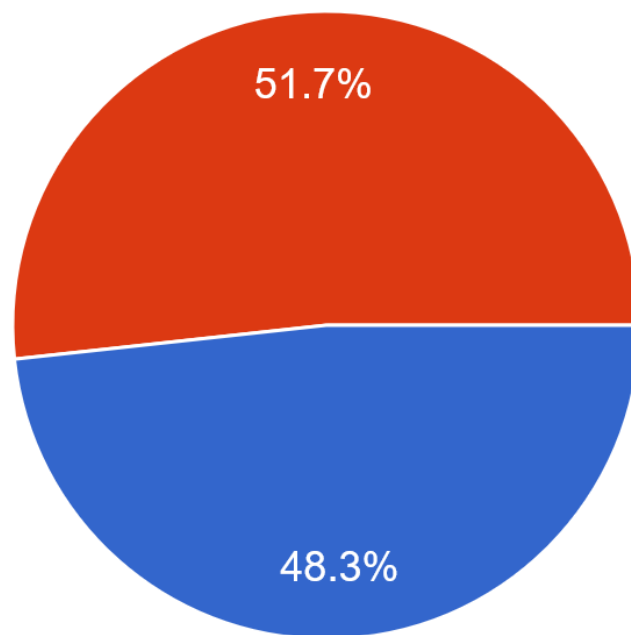
29 responses



- custom trained image classification and detection
- using language models to extract structured data from inputs (e. g. PDFs or collection labels)
- both
- I will need both in differing capacities

Which of the lessons is your top priority?

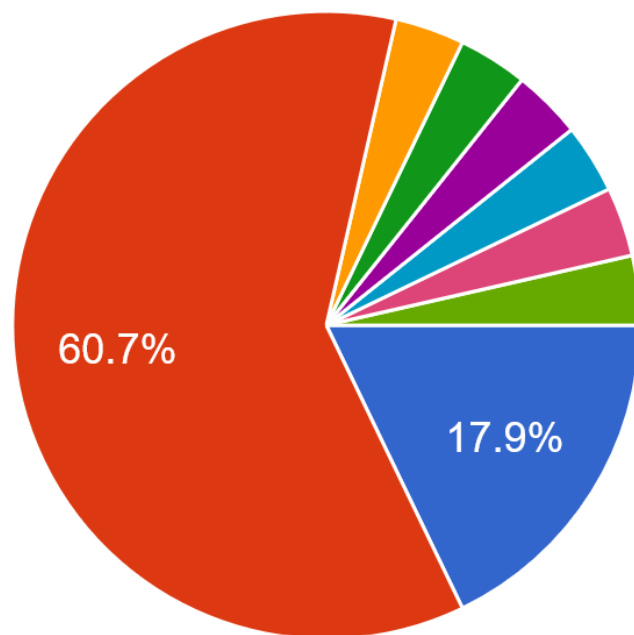
29 responses



- custom trained image classification and detection
- using language models to extract structured data from inputs (e. g. PDFs or collection labels)

Tell us about your prior experience with these topics.

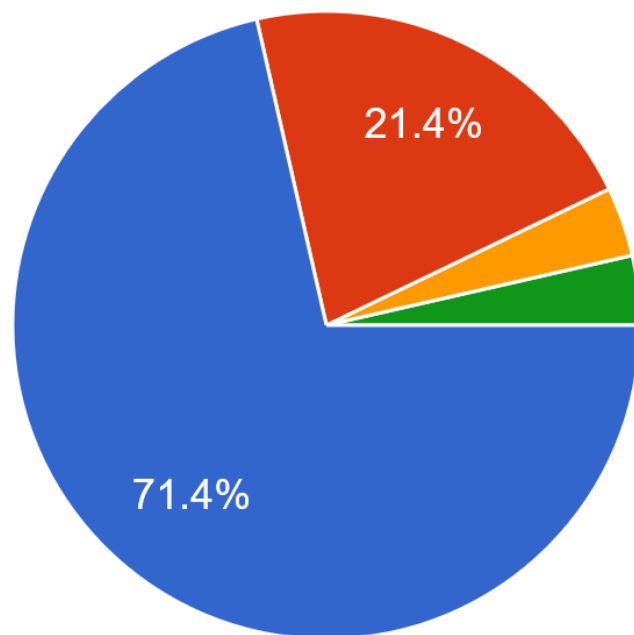
28 responses



- None
- Heard of them
- Attended a workshop
- Heard of them, read papers on them, and am exploring implementing this in...
- Working On Research Using Embedding Models
- Attended talks about the topic
- Have worked in projects that use them...
- I have read about computer vision alg...

Do you have any experience specifically with custom trained image classification and detection?

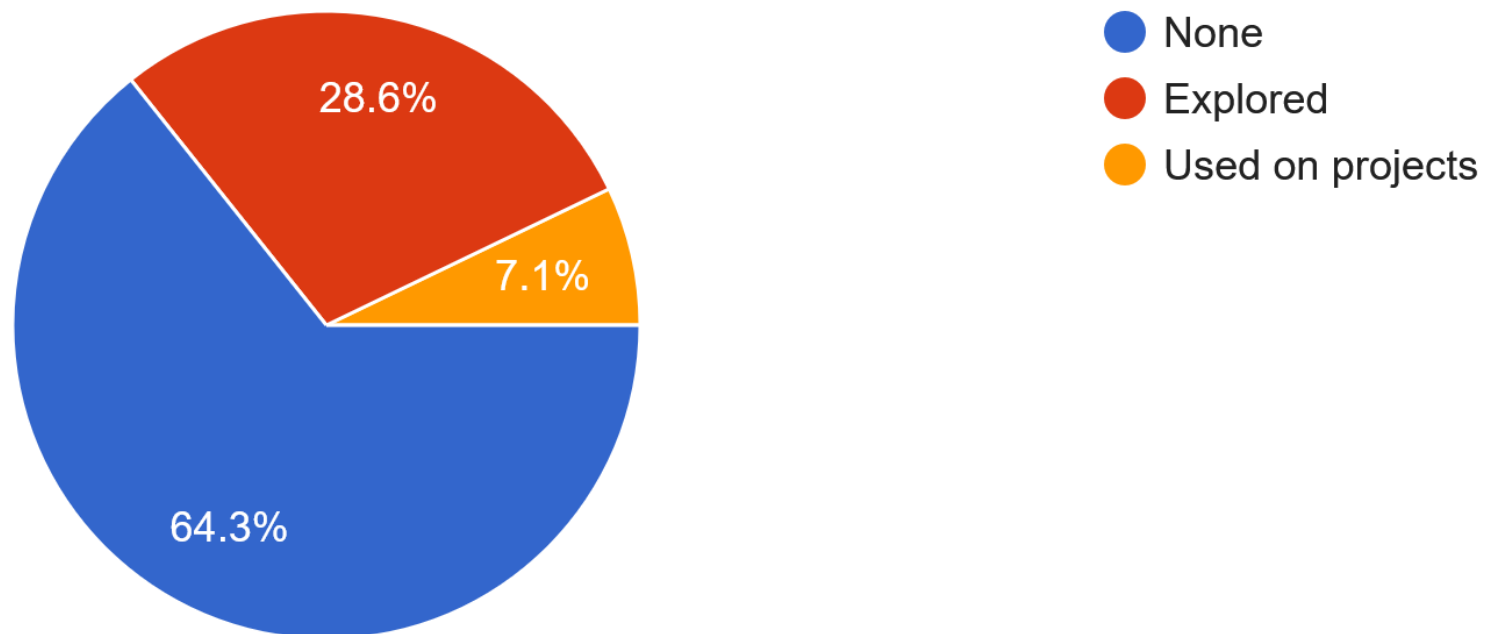
28 responses



- None
- Explored
- Used on projects
- I have used machine learning for satellite imagery habitat classifications but have never tried to do so on specimen photographs

Do you have any experience specifically with using language models to extract structured data from inputs (e. g. PDFs or collection labels)?

28 responses



Applications of ML and LLMs

Do you have any datasets (e. g. labels, PDFs, etc) and a use case where you want to apply what you learn in this workshop? If yes, please briefly describe the dataset and use case.

15 responses (volunteers to share here)?

Applications of ML and LLMs

How will this help your current career path and projects?
Tell us why this is important for you.

25 responses (volunteers to share here)?

Capacity Development across PRI and UIUC

What (if any) other related foundational computational and data science skills topics interest you that aren't covered in this workshop?

What are your current sticking / pain points?